

Deep Classification of Underwater Acoustic Signatures: A Comprehensive Analysis of the ShipsEar Dataset

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April 21, 2026

Abstract

The classification of underwater vessel noise is a critical component of passive sonar systems and maritime surveillance. This paper provides a formal overview of a classification framework utilizing the ShipsEar dataset. We explore the acoustic theory governing sound propagation in littoral zones, the mathematical representation of non-stationary signals via Mel-Frequency Cepstral Coefficients (MFCCs), and the formalization of the multi-class optimization problem. Our analysis emphasizes the physical differences between five distinct vessel categories and the cross-entropy loss framework used for model convergence.

1. Introduction and Dataset Context

The foundational data source for this system is the **ShipsEar dataset**, a comprehensive repository of underwater acoustic recordings curated by Santos-Domínguez et al. (2016). Collected along the Spanish Atlantic coast, these recordings represent real-world hydroacoustic data capturing the radiated noise of various vessels under operational conditions.

The dataset comprises 90 individual acoustic recordings, totaling approximately three hours of high-fidelity sound. The signals were sampled at $f_s = 52,734$ Hz, allowing for a wide-spectrum analysis ranging from low-frequency structural vibrations to high-frequency propeller cavitation.

2. Class Stratification and Acoustic Profiles

The system is designed to classify five mutually exclusive categories, each possessing a unique spectral signature:

- **CargoShip:** Large-scale commercial vessels characterized by massive propulsion systems, producing significant low-frequency tonals.
- **KaiYuan:** Specialized naval vessels with distinct hull designs that produce unique acoustic signatures.
- **Noise:** Ambient baseline, including waves, wind, and marine life, serving as the negative control class.
- **SpeedBoat:** High-RPM craft producing prominent high-frequency cavitation.
- **Uuv:** Unmanned Underwater Vehicles characterized by subtle, high-frequency electric motor signatures.

3. Acoustic Theory and Propagation

Underwater acoustics differ significantly from atmospheric sound. The speed of sound c in seawater is modeled by the UNESCO equation as a function of temperature (T), salinity (S), and depth (D).

$$c(S, T, D) \approx 1449.2 + 4.6T - 0.055T^2 + 1.34S - 35 + 0.016D \quad (1)$$

In shallow water zones, the sea surface and sea floor act as a waveguide, leading to severe **multipath propagation**. The received signal $x(t)$ is a sum of the direct path and various reflected paths, which introduces frequency-selective fading.

4. Feature Extraction and Signal Representation

To convert raw temporal sequences into discriminative feature vectors, we employ the Mel scale, which approximates the human ear's non-linear perception of frequency.

4.1. Mel-Frequency Transformation

The conversion from physical frequency f (in Hz) to the Mel scale m is defined by the following logarithmic relationship:

$$m = 2595 \cdot \log_{10} \left(1 + \frac{f}{700} \right) \quad (2)$$

4.2. MFCC Computation

The feature extraction pipeline involves a Discrete Cosine Transform (DCT) to decorrelate the log-energies obtained from the Mel filterbank. The k -th MFCC coefficient c_k is represented as:

$$c_k = \sum_{m=1}^M \log S_m \cdot \cos \left[\frac{\pi k}{M} \left(m - \frac{1}{2} \right) \right] \quad (3)$$

where S_m is the energy in the m -th Mel filter and M is the total number of filters.

5. Formal Problem Definition

The objective is to optimize a mapping function $f_\theta : \mathcal{X}' \rightarrow \mathcal{Y}$ parameterized by θ .

5.1. Input and Output Spaces

Let $x \in \mathcal{X}$ be a discrete-time sequence. The feature extraction maps x to a feature space $\mathcal{X}' \subset \mathbb{R}^{d \times \tau}$, where d is the number of filters and τ is the number of time frames.

5.2. Loss Function and Optimization

For a multi-class problem with $C = 5$ classes, we minimize the **Categorical Cross-Entropy Loss**. Given a ground-truth one-hot vector y and the predicted probability distribution $\hat{y}$, the loss $\mathcal{L}$ is defined as:

$$\mathcal{L}(y, \hat{y}) = - \sum_{i=1}^C y_i \ln \hat{y}_i \quad (4)$$

The predicted probability for each class is calculated via the **Softmax** function:

$$\hat{y}_i = \frac{e^{z_i}}{\sum_{j=1}^C e^{z_j}} \quad (5)$$

6. Evaluation and Conclusion

The performance is evaluated using the **F1-score**, critical for handling potential class imbalances in underwater datasets:

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (6)$$

This framework establishes a robust mathematical foundation for identifying maritime vessels in complex, high-noise underwater environments.